Kinvecta

A KINVECTA SERIES

How Large Language Models Work

From math to transformers

EPISODE 03 · MATRICES AS MACHINES



Kinvecta

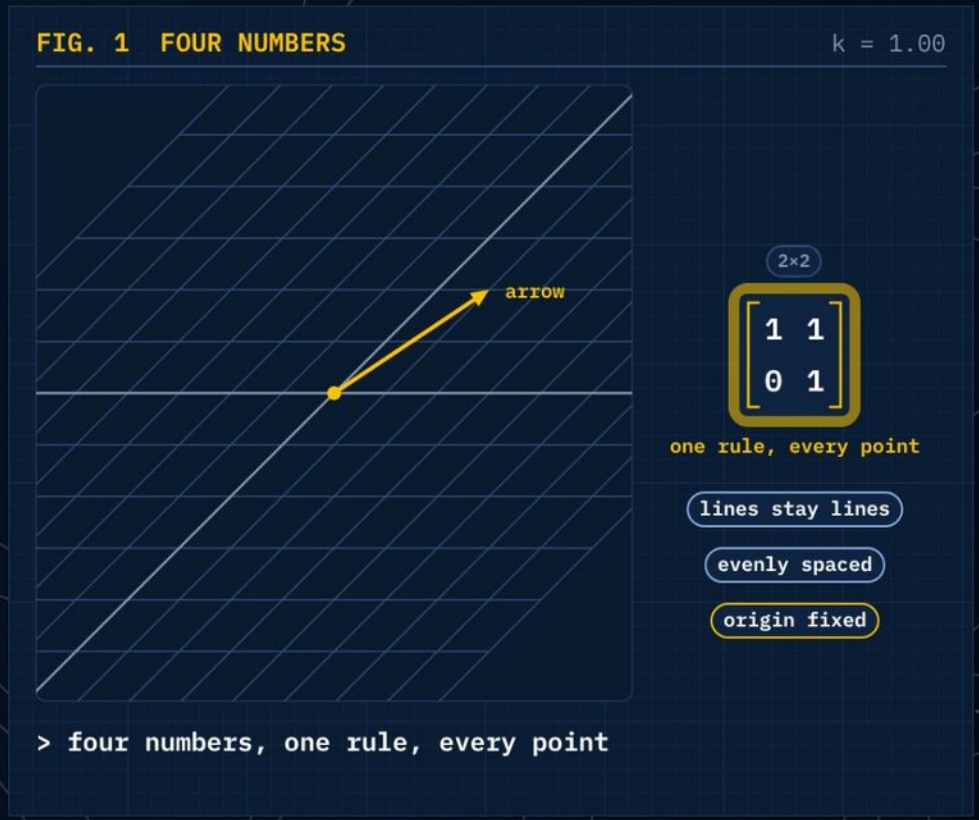
HOW LARGE LANGUAGE MODELS WORK · EPISODE 3

Four numbers. Every point moves.

A grid of numbers is a rule that moves every vector at once. That rule is a matrix.

15-minute lesson

Companion notebook



MARK

WHY THIS MATTERS

A matrix is a machine

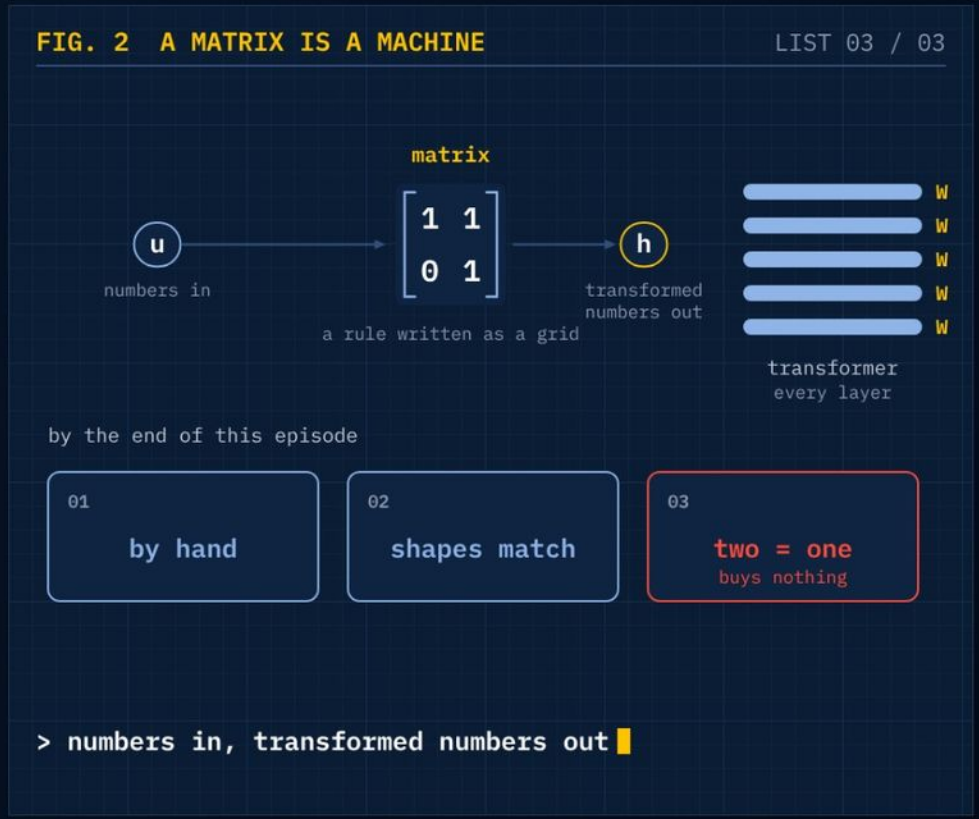
Numbers in, transformed numbers out, by a rule written as a grid.

01 Every layer of every transformer is one of these

02 Multiply matrices by hand

03 Know why shapes must match

04 See why two stacked layers buy nothing



3Blue1Brown, “Linear transformations and matrices,” Essence of linear algebra, ch. 3 (2016). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §§6.1, 6.3.

EPISODE 3 · MATRICES AS MACHINES

Matrices as machines

Four parts, one grid of numbers, and a notebook with every number.

01 A matrix as a rule that moves vectors

02 The product, and the shape rule

03 The linear layer: h = Wu + β

04 Stacking tokens into rows, plus a myth about depth



Companion notebook: “Episode 03 - Matrices” (Colab), linked in the description. Sources on every slide.

PART 1 · WHAT “LINEAR” MEANS

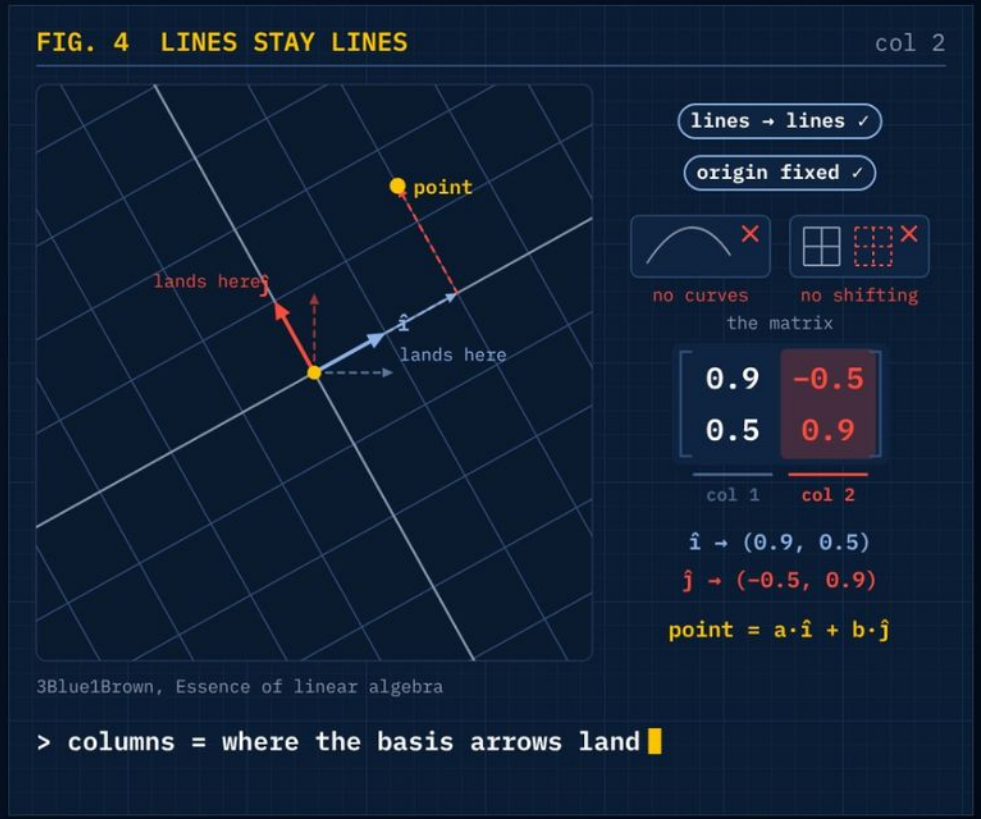
Lines stay lines

The informal test: straight lines stay straight, and the origin stays put.

01 No curves, no shifting

02 The first column: where the first basis arrow lands

03 The second column: where the second lands



3Blue1Brown, “Linear transformations and matrices,” Essence of linear algebra, ch. 3 (2016).

PART 1 · MATRIX × VECTOR

Scale the columns, add

Two readings of the same arithmetic.

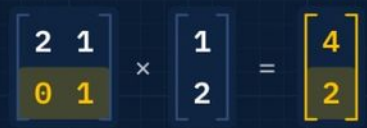
01 Columns view: a weighted mix of the columns

02 Rows view: each output is a dot product

03 Component i of Wu = row i of W · u

FIG. 5 MATRIX × VECTOR — ROWS

illustrative



a second way: one dot product per row

row 1 · u = 2·1 + 1·2 = 4

row 2 · u = 0·1 + 1·2 = 2

$(W\,u)_i = \text{row}_i \cdot u$

component i = row i dotted with u

episode 2: multiply matching entries, add

done once per row

> columns view or rows view: same number

3Blue1Brown, “Linear transformations and matrices,” Essence of linear algebra, ch. 3 (2016). Dot product: Episode 2. Numbers in the figure are illustrative.

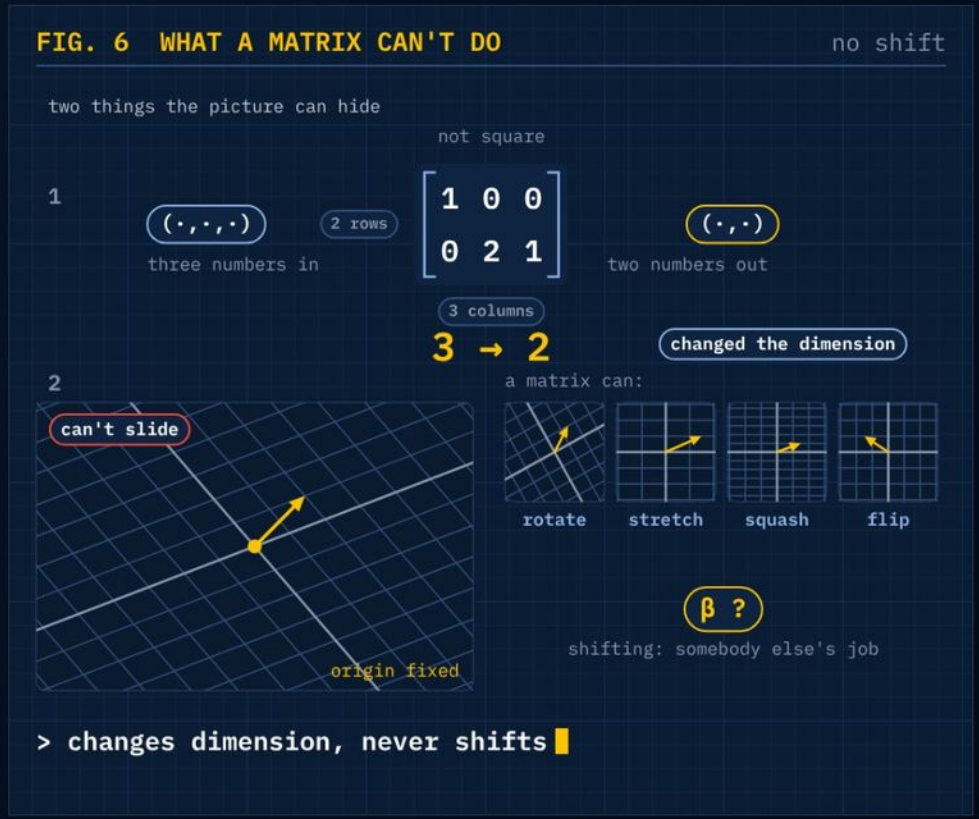
What a matrix can’t do

It can change dimension. It can never shift.

01 A 2×3 matrix: three numbers in, two out

02 Rotate, stretch, squash, flip, yes

03 Slide sideways, no: the origin stays fixed



3Blue1Brown, “Linear transformations and matrices,” Essence of linear algebra, ch. 3 (2016). All weights and word vectors are hand-chosen teaching values, not learned.

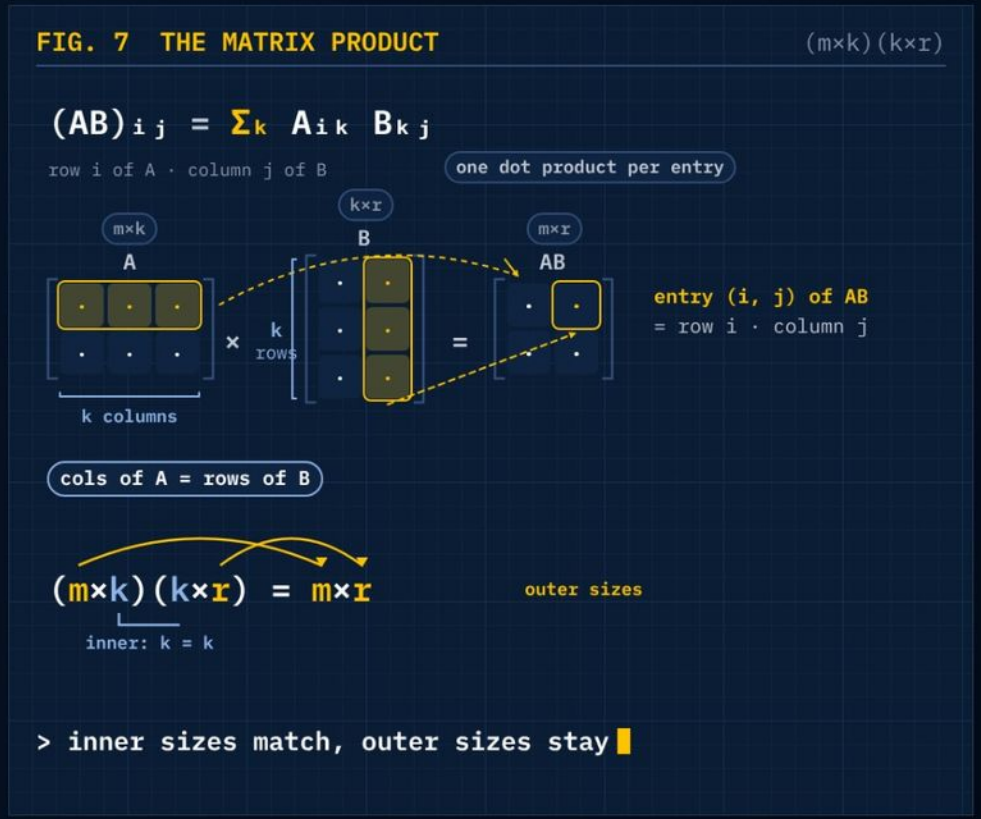
One dot product per entry

$(AB)_{ij} = \Sigma_k A_{ik} B_{kj}$: row i of A dotted with column j of B.

01 A's columns must equal B's rows

02 Inner sizes match, outer sizes stay

03 (m×k)(k×r) = m×r



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose).

PART 2 · BY HAND, ENTRY 1

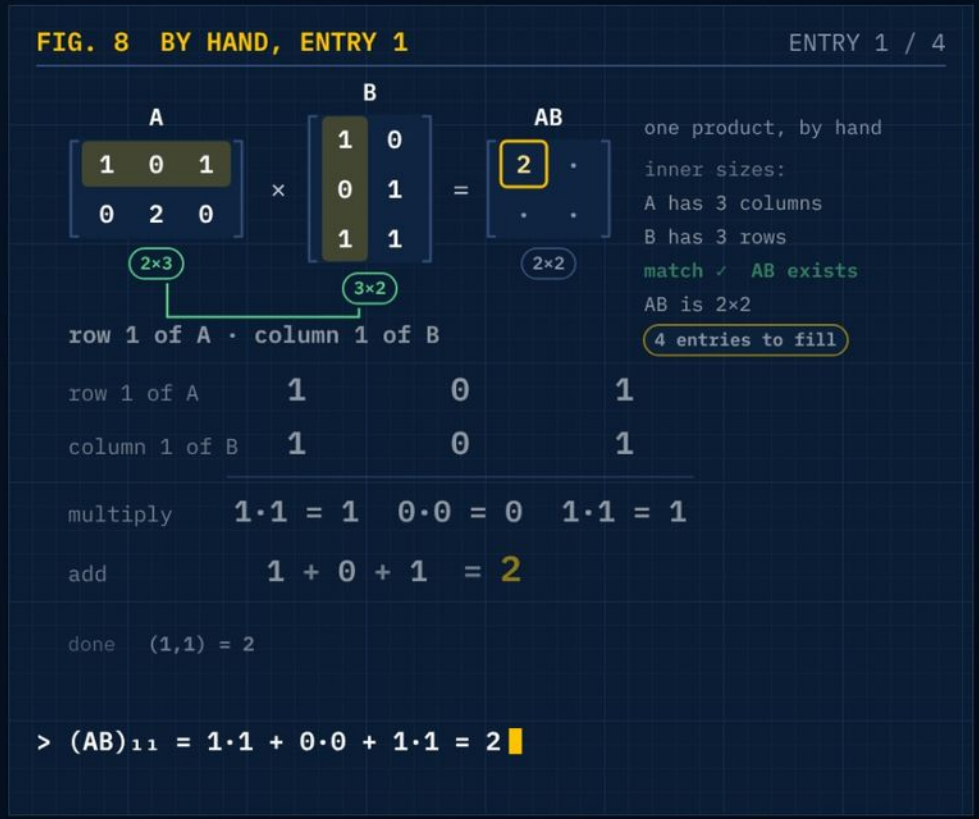
Let’s do one slowly

A = [[1, 0, 1], [0, 2, 0]] · B = [[1, 0], [0, 1], [1, 1]]

01 A is 2×3, B is 3×2: inner sizes match, so AB is 2×2

02 Entry (1, 1): row 1 of A · column 1 of B

03 1·1 + 0·0 + 1·1 = 2



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose). All weights and word vectors are hand-chosen teaching values, not learned.

MARK

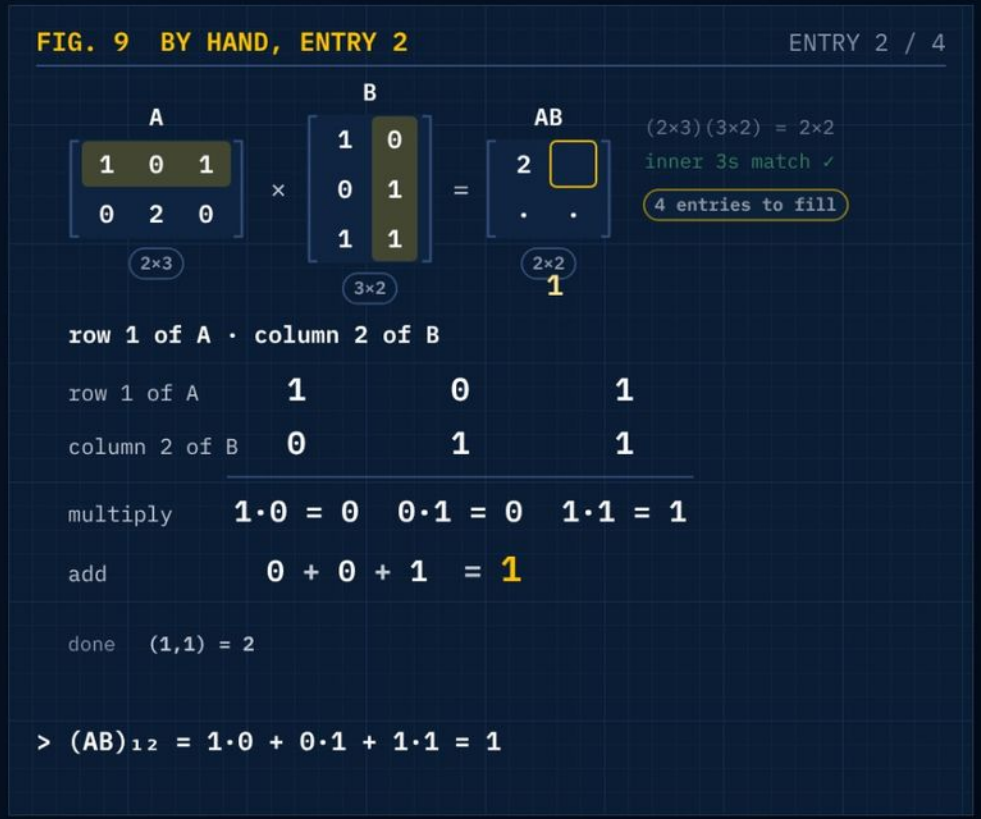
PART 2 · BY HAND, ENTRY 2

Same row, next column

Row 1 of A is (1, 0, 1). Column 2 of B is (0, 1, 1).

01 Entry (1, 2): row 1 of A · column 2 of B

02 1·0 + 0·1 + 1·1 = 1



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose). All weights and word vectors are hand-chosen teaching values, not learned.

PART 2 · BY HAND, ROW 2

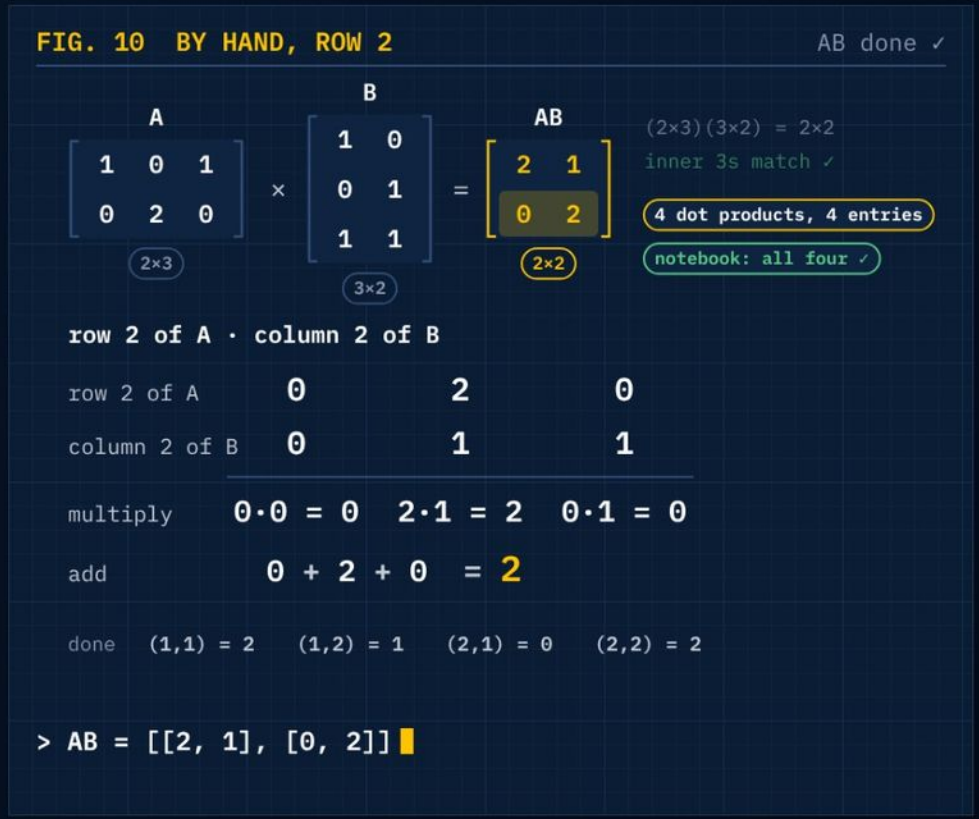
Two more, and it's done

Row 2 of A is (0, 2, 0).

01 Entry (2, 1): 0·1 + 2·0 + 0·1 = 0

02 Entry (2, 2): 0·0 + 2·1 + 0·1 = 2

03 AB = [[2, 1], [0, 2]]: four dot products, four entries



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose). All weights and word vectors are hand-chosen teaching values, not learned.

Does the product exist?

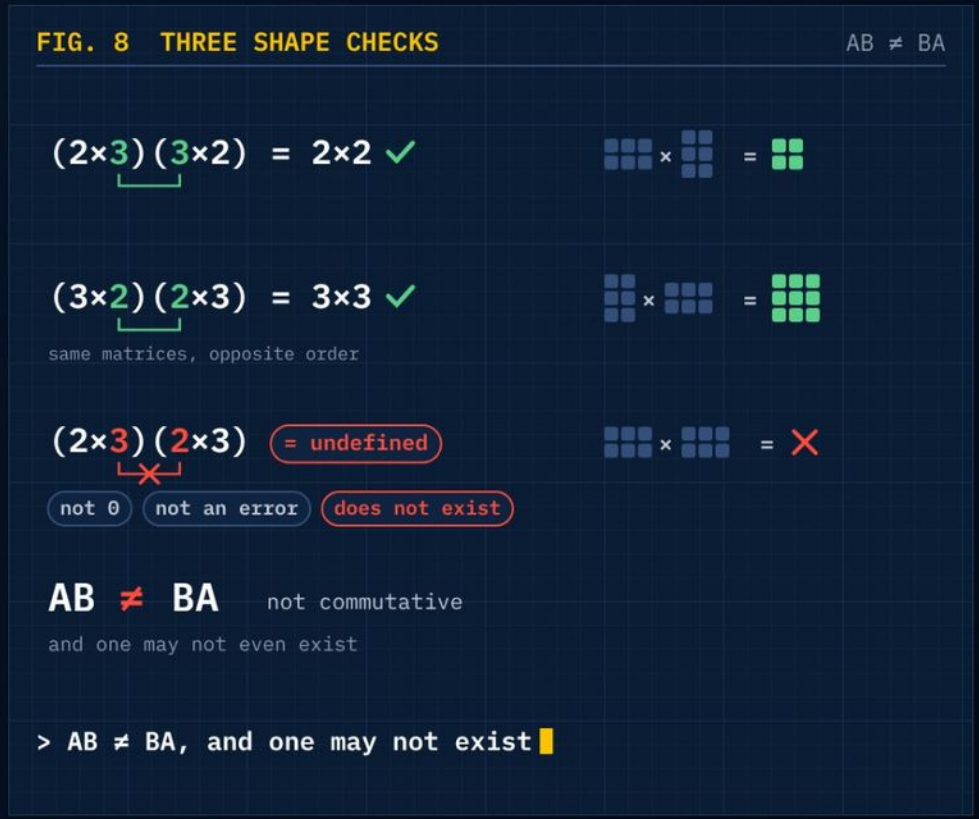
Same two matrices, opposite order, different shape, or no product at all.

01 (2×3)(3×2) = 2×2

02 (3×2)(2×3) = 3×3

03 (2×3)(2×3): undefined

04 AB ≠ BA: not commutative



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose).

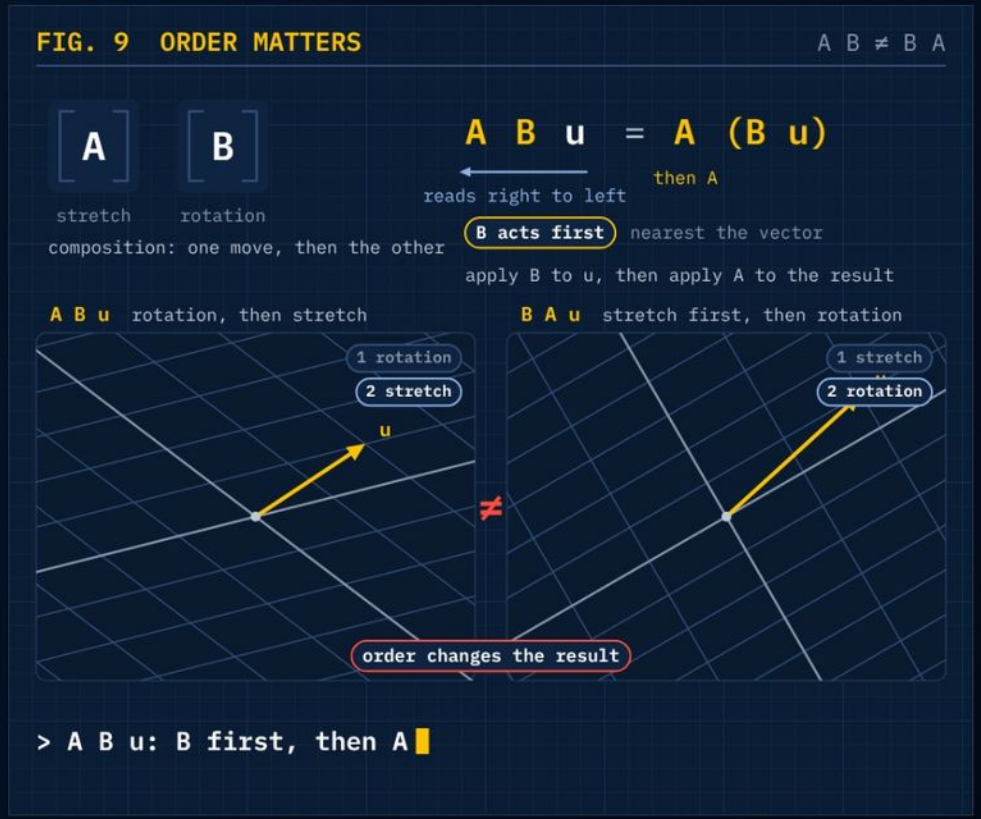
Composition, right to left

AB u means: apply B to u, then apply A to the result.

01 The matrix nearest the vector acts first

02 Rotate then stretch ≠ stretch then rotate

03 Order changes the result



3Blue1Brown, “Matrix multiplication as composition,” Essence of linear algebra, ch. 4 (2016).

PART 3 · THE LINEAR LAYER

h = Wu + β

The one line every neural network is built from.

01 u: the input, d numbers

02 W: the weights, d_out × d

03 β: the bias, d_out numbers

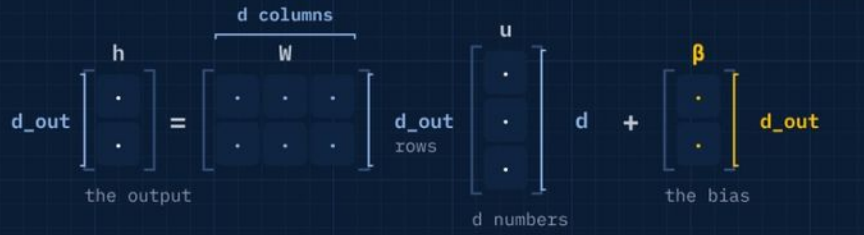
04 Goodfellow et al. write it z = Wx + b

FIG. 10 THE LINEAR LAYER — DL §6.3

the one line every neural network is built from

h = W u + β

d in → d_out out



why β and not b?

b = episode 2 vector · x, y taken

Goodfellow, Bengio & Courville (2016)
Deep Learning · §6.3

z = W x + b "affine" same idea, our letters

> h = W u + β

Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §§6.1, 6.3. Our letters differ because x, y and b are reserved in this series.

PART 3 · WHAT B DOES

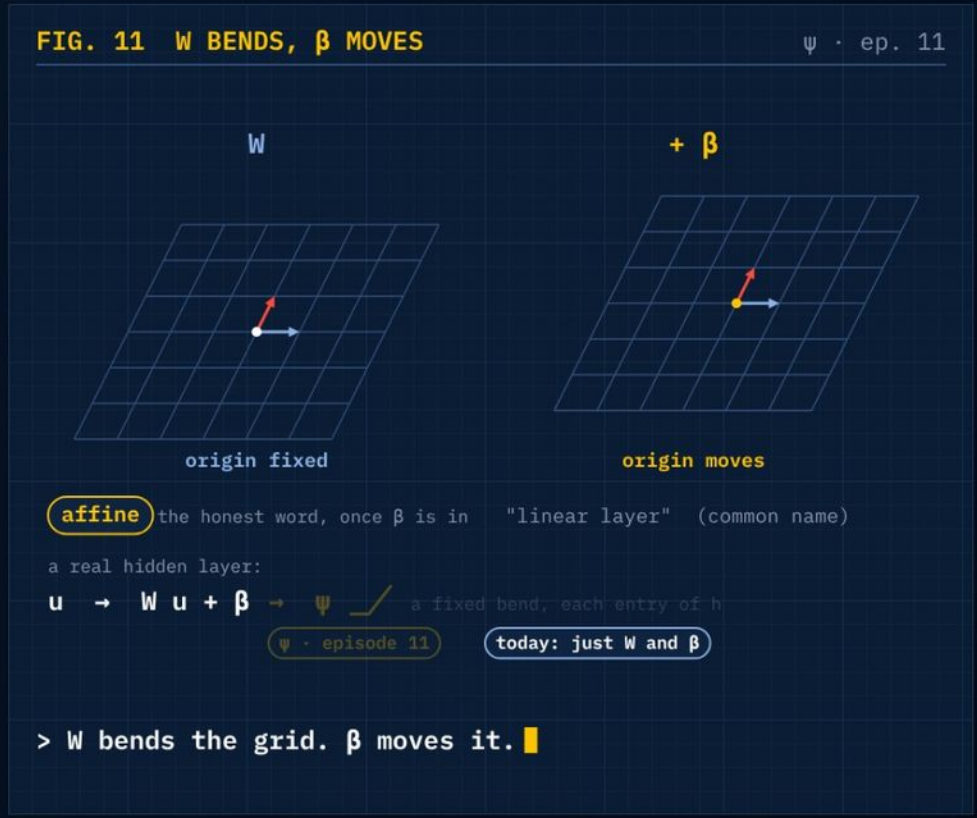
W bends the grid. β moves it.

A linear map can't shift. The bias adds the same vector to every output.

01 With β ≠ 0 the map is affine (the honest word)

02 "Linear layer" is the common name, and fine

03 A hidden layer adds a fixed bend ψ after this: Episode 11



3Blue1Brown, "Linear transformations and matrices," Essence of linear algebra, ch. 3 (2016). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §§6.1, 6.3.

PART 3 · A PICTURE TO KEEP

The mixing board

Three input strips, two output buses, a dial at every crossing.

01 The grid of dials is W

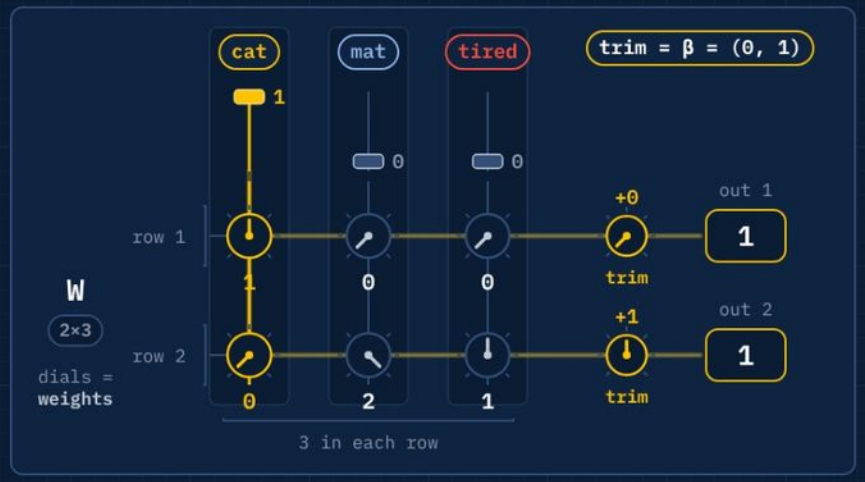
02 Each output bus is a weighted sum of the inputs

03 A trim knob per bus is β (a real board has none)

04 Here the dials are set by hand; a model learns them: Episode 7

FIG. 12 THE MIXING BOARD

$h = W u + \beta$



the trim is the bias; a real board has no such knob

dials: set by hand here | learned in a model (ep. 7)

> each output = a weighted sum of inputs

Kinvecta analogy, built for this series. All weights and word vectors are hand-chosen teaching values, not learned.

MATH LAB · STEP 1

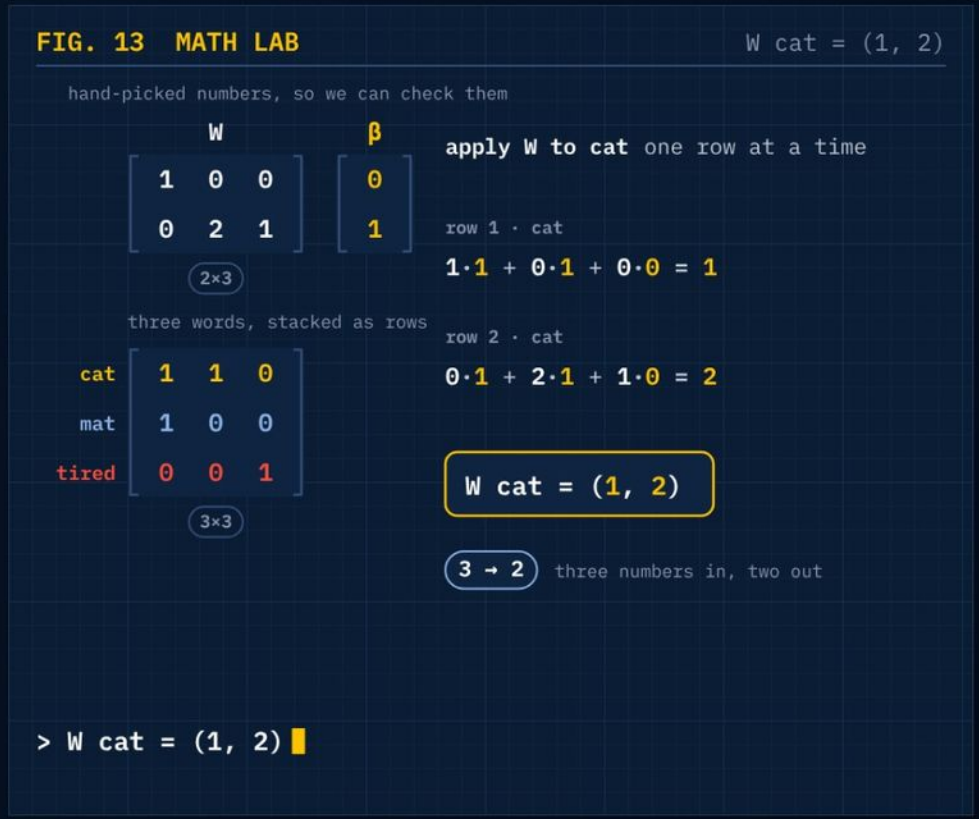
Apply W to cat

`W = [[1, 0, 0], [0, 2, 1]], β = (0, 1);`
`rows cat, mat, tired.`

01 Row 1 · cat = 1·1 + 0·1 + 0·0 = 1

02 Row 2 · cat = 0·1 + 2·1 + 1·0 = 2

03 W cat = (1, 2): three numbers in, two out



All weights and word vectors are hand-chosen teaching values, not learned.

MATH LAB · STEP 2

Add the bias

W mat = (1, 0) · W tired = (0, 1) · β = (0, 1)

01 cat → (1, 3)

02 mat → (1, 1)

03 tired → (0, 2)

04 Three outputs, two numbers each, one shared shift



All weights and word vectors are hand-chosen teaching values, not learned.

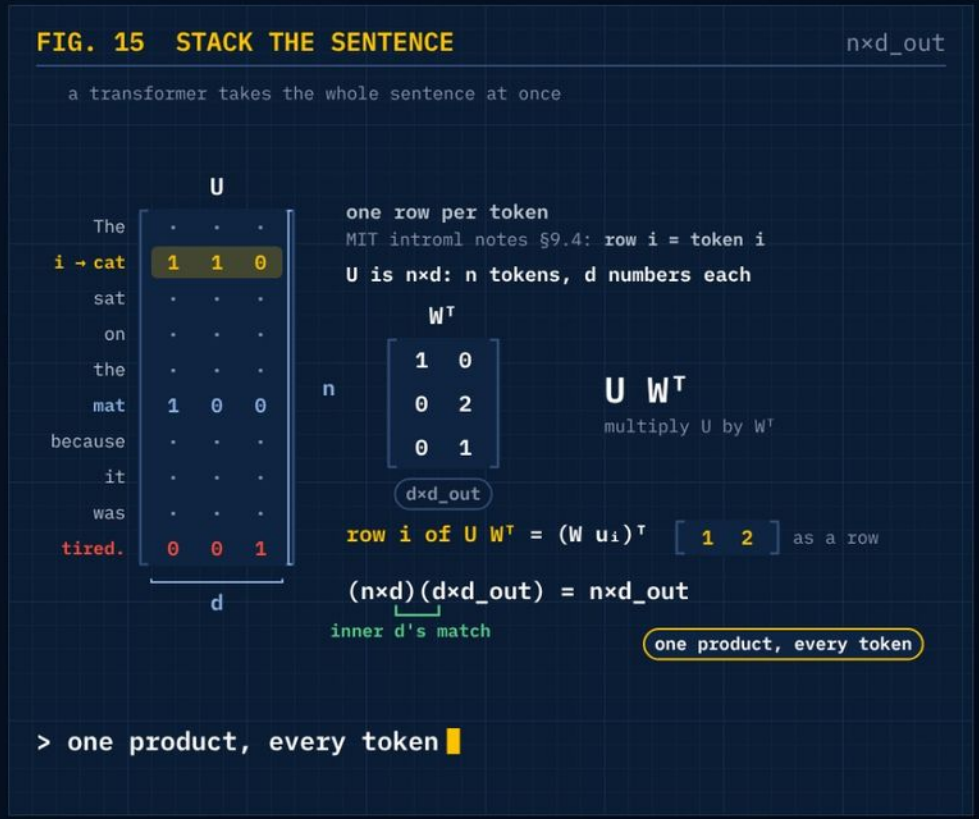
Stack the tokens as rows

U is n × d: one row per token. The MIT notes use the same convention.

01 Row i of U is token i

02 Batched layer: U W^T, then β on every row

03 (n×d)(d×d_out) = n×d_out



MIT introml, "Transformers" notes, §9.4 (row convention). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose). (transpose).

PART 4 · ONE PRODUCT, ROW ONE BY HAND

U W^T, entry by entry

W^T = [[1, 0], [0, 2], [0, 1]] (3×2) · U
= the three words stacked (3×3)

01 Inner threes match: the result is 3×2

02 Row 1 · column 1: 1·1 + 1·0 + 0·0 = 1

03 Row 1 · column 2: 1·0 + 1·2 + 0·1 = 2

FIG. 19 U W^T, ROW ONE BY HAND — row 1 = (1, 2)

	U				W^T			U W^T	
cat	1	1	0		1	0		1	2
mat	1	0	0	×	0	2	=	·	·
tired	0	0	1		0	1		·	·
	3×3				3×2			3×2	

watch it happen

(3×3)(3×2) = 3×2

inner 3s match

= W cat ✓

exactly as before

row 1 of U · column 2 of W^T

cat (row 1)	1	1	0
column 2 of W^T	0	2	1

multiply 1·0 = 0 1·2 = 2 0·1 = 0

add 0 + 2 + 0 = 2

done (1,1) = 1 (1,2) = 2 row 1 = (1, 2)

> row 1 of U W^T = (1, 2) = W cat

MIT introml, “Transformers” notes, §9.4 (row convention). All weights and word vectors are hand-chosen teaching values, not learned.

PART 4 · ONE PRODUCT

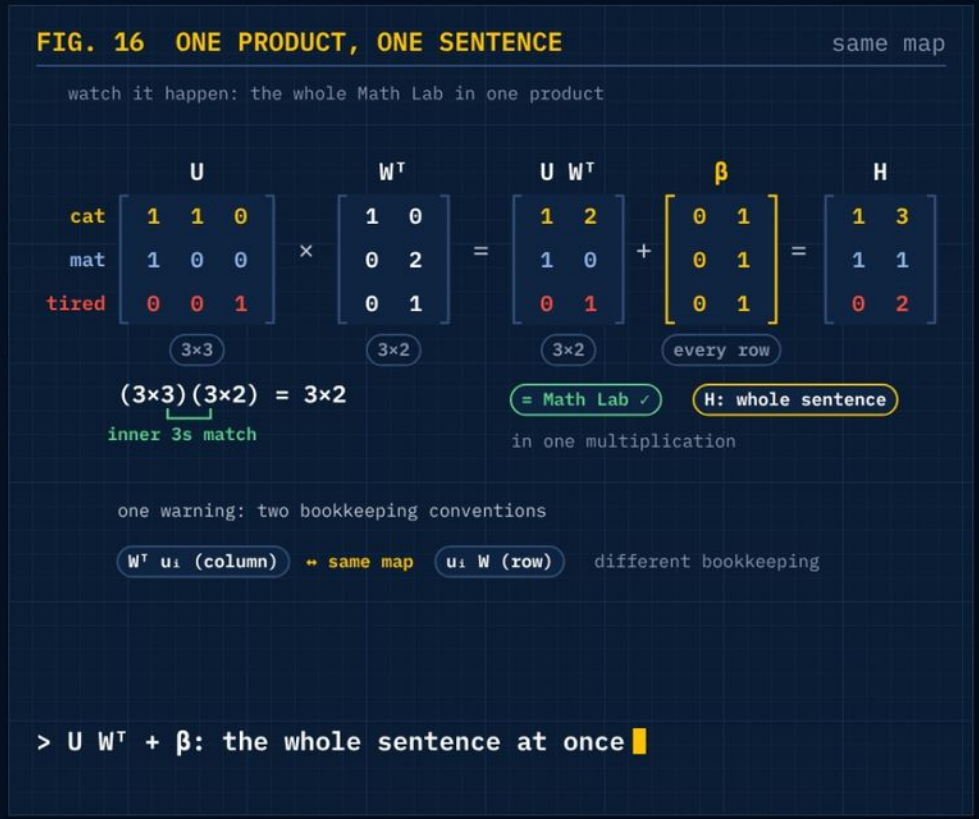
The whole sentence at once

Row 1 is (1, 2): W applied to cat, exactly as before.

01 Row 2, mat: (1, 0) · Row 3, tired: (0, 1)

02 Exactly the Math Lab results, in one multiplication

03 Column form $W^T u_i$ or row form $u_i W$: same map



MIT introml, “Transformers” notes, §9.4 (row convention). All weights and word vectors are hand-chosen teaching values, not learned.

PART 4 · PREVIEW

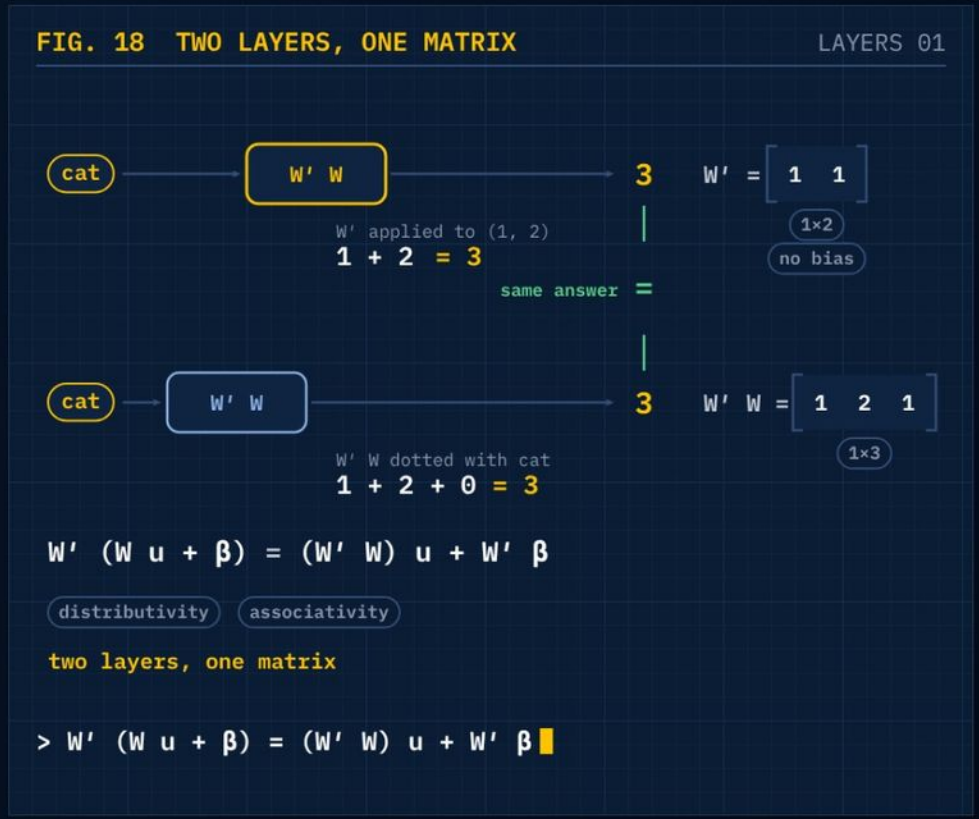
Two layers, one matrix

W' = [1, 1] after W, no bias. Two routes to the same number.

01 W' (W cat) = W' (1, 2) = 3

02 W'W = [1, 2, 1], and [1, 2, 1] · cat = 3

03 W'(Wu + β) = (W'W)u + W'β



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 2 §2.2 (product, shape rule, transpose). (distributivity, associativity). All weights and word vectors are hand-chosen teaching values, not learned.

MYTH CHECK

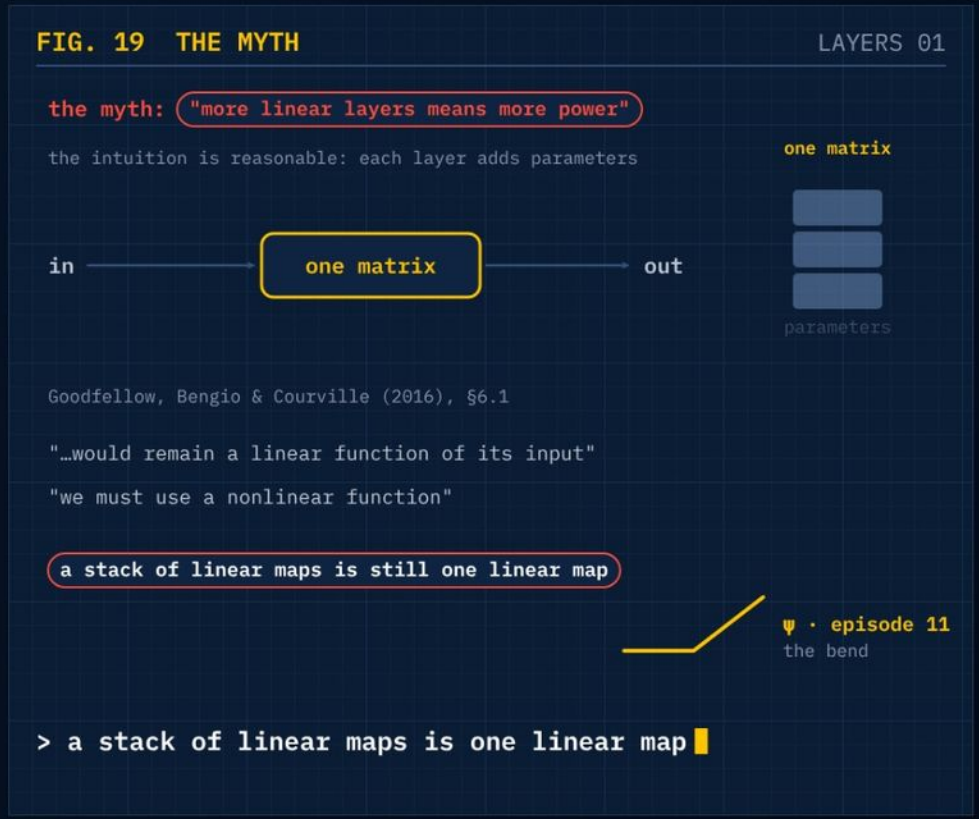
“More linear layers means more power”

Reasonable intuition. Wrong conclusion.

01 A linear layer keeps the whole network linear (Goodfellow §6.1)

02 “We must use a nonlinear function”

03 A stack of linear maps is still one linear map



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §§6.1, 6.3.

MYTH CHECK · THE FINE PRINT

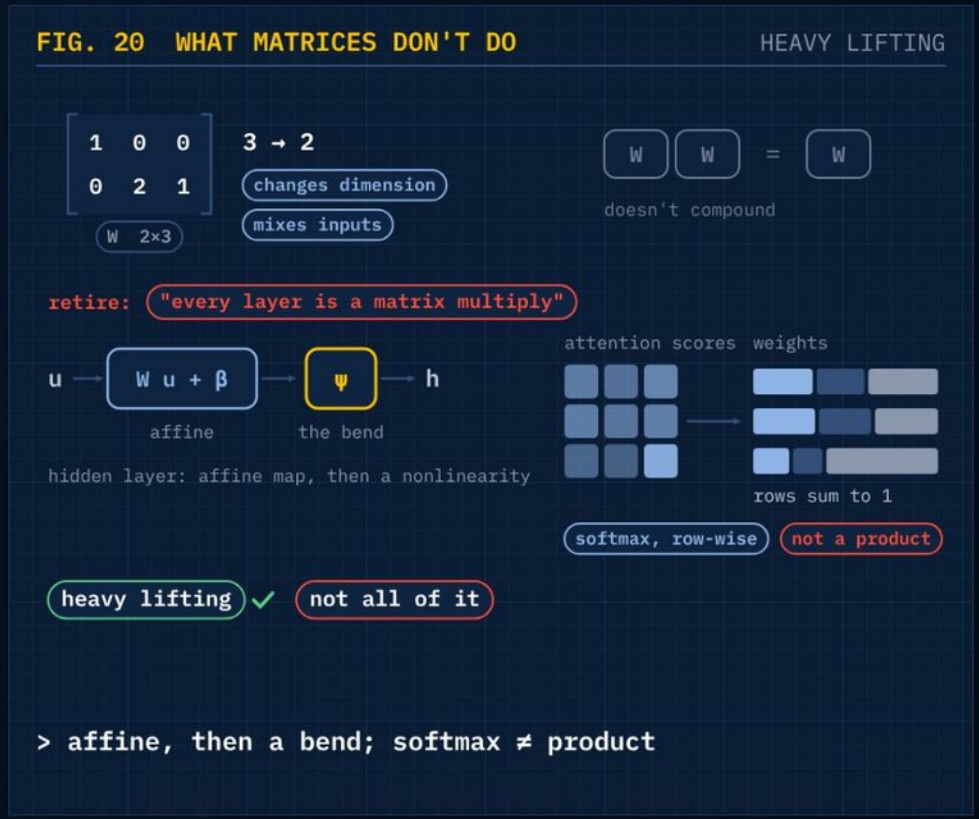
Matrices do the heavy lifting

Linear layers change dimension and mix inputs. They just don't compound.

01 "Every layer is a matrix multiply" is overstated

02 A hidden layer: affine map, then a nonlinearity

03 Attention's softmax is applied per row, not a product



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 6 §§6.1, 6.3. MIT introml, "Transformers" notes, §9.4 (row convention).

Kinvecta

RECAP

Five ideas

Everything in the series leans on this episode.

01 Columns say where the basis arrows land

02 A product is composition; inner sizes must match

03 h = Wu + β: β is what lets it shift

04 Rows for tokens; two linear layers are one



All sources are listed in the description.

MARK

YOUR TURN

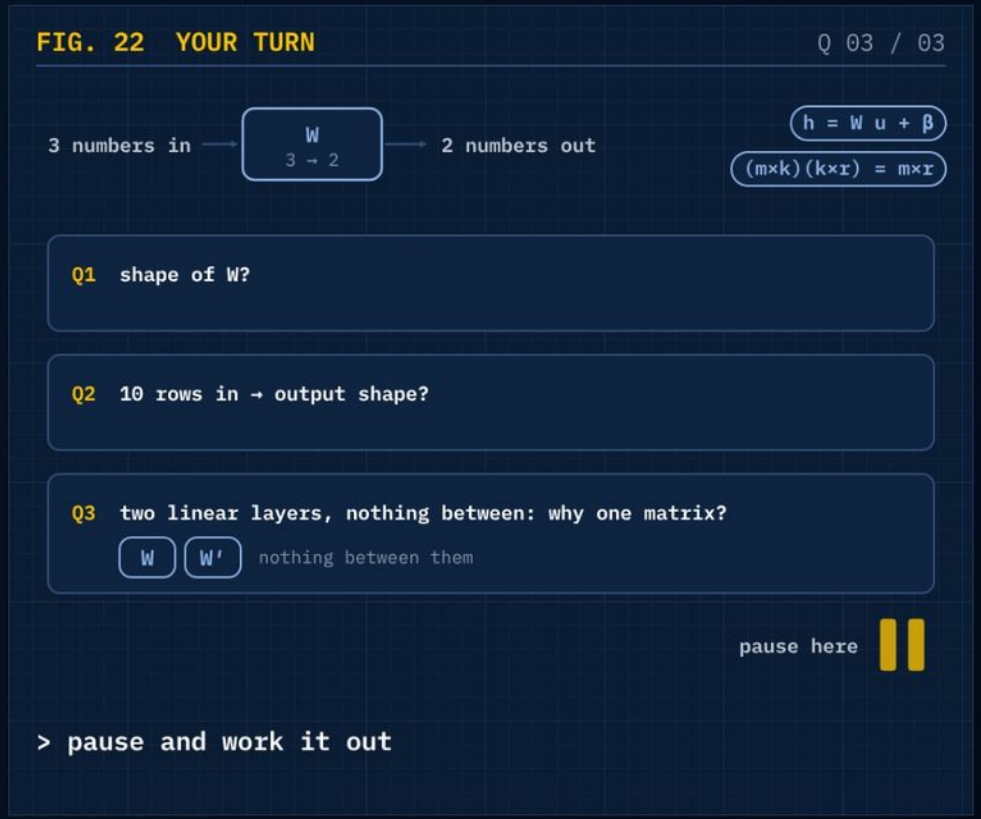
One question to take with you

A layer takes 3-number inputs and returns 2-number outputs.

01 What shape is W?

02 10 token vectors stacked as rows: output shape?

03 Two linear layers, nothing between: why one matrix?



All weights and word vectors are hand-chosen teaching values, not learned.

YOUR TURN

The answer

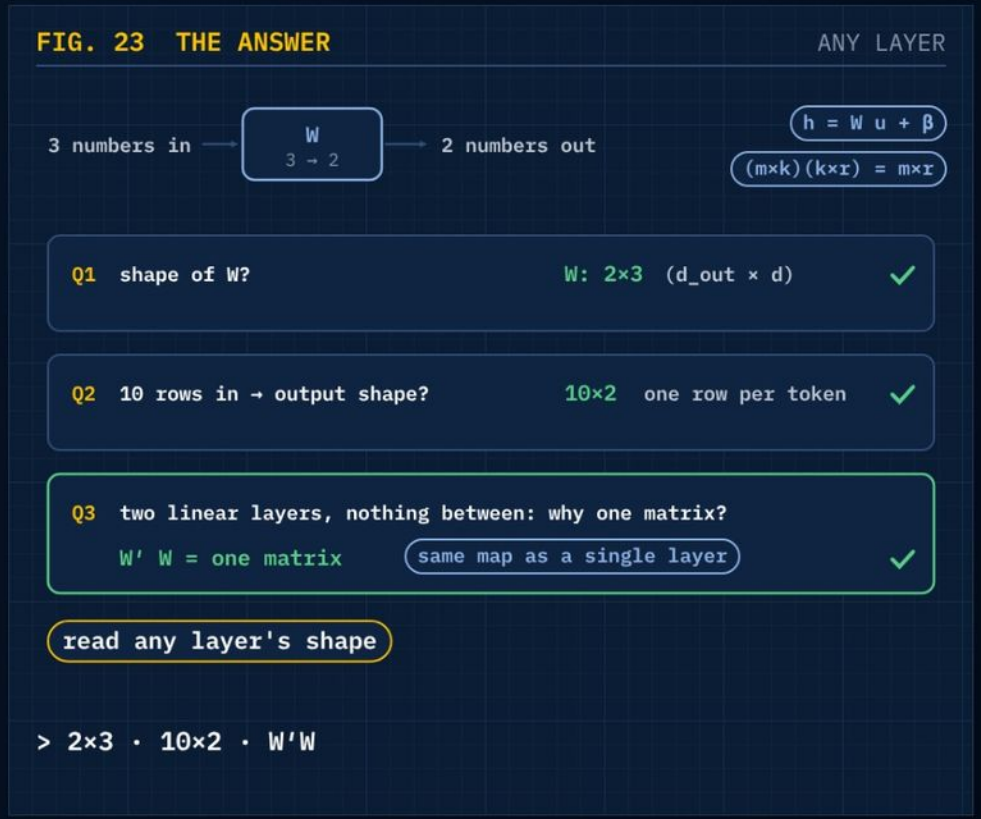
Shapes first, then the collapse.

01 W is 2×3: d_out rows, d columns

02 The batched output is 10×2

03 The two matrices multiply into one

04 You can read the shape of any layer in any model



All weights and word vectors are hand-chosen teaching values, not learned.

Kinvecta

TRY THIS WEEK

Three things to try.

01

Multiply a 2×3 and a 3×2 matrix both ways. Check shapes first

02

In the notebook, change one dial in W and watch the outputs

03

Build your own collapse: W'W, then confirm it on cat

04

Say out loud what each shape tag means

THE MISTAKE TO AVOID

Counting a stack of linear layers as depth. Without a bend between them, it’s one layer wearing a costume.

Companion notebook and every source in the description.

MARK

Kinvecta

NEXT TIME

Episode 4: Probability, entropy, perplexity.
Numbers become probabilities. Then we measure surprise.

In the comments: which trips you up more, the shape rule, or that order matters?

How Large Language Models Work · subscribe for Episode 4



MARK